

WORCESTER, MA, USA

WMO: 725100

Lat: 42.271N

Lon: 71.873W

Elev: 305

StdP: 97.72

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94746

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-14.0	-24.8	0.4	-14.9	-22.3	0.5	-12.1	13.9	-7.1	12.6	-4.9	6.2	270	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	30.0	21.9	28.6	20.9	27.2	20.1	23.4	27.6	22.5	26.3	21.7	25.2	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.4	25.3	21.3	16.6	24.4	20.6	15.8	23.7	70.9	27.6	67.5	26.2	64.7	25.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.1	8.8	-20.0	32.5	2.7	1.4	-22.0	33.5	-23.5	34.4	-25.0	35.2	-27.0	36.2		
(5)			-20.7	25.0	2.6	1.2	-22.5	25.9	-24.0	26.6	-25.5	27.3	-27.4	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.1	-2.6	1.5	8.0	13.7	18.6	21.6	20.8	16.9	10.5	4.7	-0.7		
	DBStd	10.02	6.00	5.51	5.35	4.89	4.44	3.81	2.91	2.86	3.78	4.36	4.95	5.17		
	HDD10.0	1720	436	354	271	94	13	1	0	0	1	47	171	332		
	HDD18.3	3645	694	587	523	312	157	43	6	10	70	244	409	589		
	CDD10.0	1407	1	1	6	34	127	257	361	335	209	63	12	2		
	CDD18.3	291	0	0	0	2	13	50	108	87	28	2	0	0		
	CDH23.3	1798	0	0	3	27	118	333	699	476	136	7	0	0		
	CDH26.7	398	0	0	1	7	25	77	169	94	25	1	0	0		
(14) Wind	WSAvg	4.5	5.1	5.2	5.1	4.8	4.4	4.1	3.9	3.7	3.9	4.4	4.7	5.0		
(15) Precipitation	PrecAvg	1193	89	83	105	98	105	100	97	96	107	102	109	99		
	PrecMax	1820	284	213	202	223	252	309	201	204	334	259	264	250		
	PrecMin	812	23	6	19	19	30	8	19	26	18	18	17	19		
	PrecStd	198	55	39	43	47	49	66	46	43	64	50	52	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	13.8	21.3	27.3	29.4	31.1	32.1	31.1	29.5	24.0	18.8	15.9		
		MCWB	12.2	10.2	13.0	15.8	19.2	22.2	23.5	22.7	21.3	16.9	14.3	13.3		
	2%	DB	10.6	10.3	15.5	21.9	26.7	28.9	30.0	29.0	26.7	21.4	16.4	12.1		
		MCWB	8.7	7.0	9.3	13.0	17.3	21.1	22.3	21.8	19.8	16.7	13.3	10.0		
	5%	DB	7.1	7.5	12.0	18.7	23.9	27.0	28.6	27.5	24.7	19.3	14.2	9.5		
		MCWB	5.3	4.3	7.5	11.5	16.1	19.8	21.3	20.7	18.9	15.4	11.3	7.6		
	10%	DB	3.9	4.8	9.1	15.9	21.5	25.1	27.2	26.1	22.8	17.4	12.2	6.9		
		MCWB	1.9	2.1	5.5	9.8	14.9	18.5	20.5	19.8	18.1	13.9	9.4	4.6		
	0.4%	WB	13.0	12.0	14.1	17.4	20.9	23.7	24.7	24.0	22.8	20.0	16.7	13.6		
		MCDB	13.6	13.4	20.1	23.2	27.2	28.8	29.7	28.6	26.0	21.6	17.7	14.5		
	2%	WB	9.1	7.9	11.2	15.0	19.2	22.4	23.6	23.0	21.5	18.1	14.3	10.5		
		MCDB	10.0	9.4	14.1	20.4	23.8	27.1	28.0	26.6	24.5	20.0	15.8	11.8		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	5.3	4.7	8.3	13.0	17.8	21.3	22.7	22.2	20.6	16.2	12.2	7.8		
		MCDB	6.9	7.0	11.0	16.7	22.2	25.2	26.6	25.4	23.2	18.7	13.6	9.1		
	10%	WB	2.1	2.3	5.7	10.8	16.1	20.1	21.8	21.4	19.4	14.3	9.9	4.9		
		MCDB	3.6	4.4	8.8	14.5	19.9	23.3	25.5	24.4	22.0	16.6	12.0	6.7		
	5% DB	MDBR	7.4	7.9	8.4	9.6	9.7	9.1	8.8	8.5	8.5	8.2	7.5	6.8		
		MCDBR	10.3	10.4	12.2	13.6	12.9	10.9	10.0	9.7	10.0	10.1	9.6	9.8		
(36) Mean Daily Temperature Range	5% WB	MCWBR	9.8	8.3	8.1	7.4	6.8	5.8	4.6	4.5	5.0	6.4	7.8	9.2		
		MCDBR	10.4	10.0	11.4	12.6	11.7	10.0	9.2	8.6	8.5	9.3	9.3	9.7		
	5% WB	MCWBR	10.2	8.9	8.9	8.4	7.0	6.0	4.8	4.7	5.5	7.4	8.7	9.8		
		MCDBR	10.2	8.9	8.9	8.4	7.0	6.0	4.8	4.7	5.5	7.4	8.7	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.288	0.315	0.359	0.394	0.433	0.450	0.420	0.359	0.322	0.297	0.280		
	taud		2.475	2.460	2.473	2.388	2.320	2.234	2.195	2.266	2.418	2.544	2.572	2.562		
	Ebn at Noon		879	925	935	910	881	843	824	839	873	875	853	851		
	Edh at Noon		73	87	97	114	126	138	142	128	102	80	67	62		
(44) All-Sky Solar Radiation	RadAvg		1.81	2.67	3.67	4.68	5.17	5.61	5.88	5.28	4.29	2.81	1.95	1.49		
	RadStd		0.12	0.22	0.26	0.50	0.58	0.55	0.35	0.27	0.31	0.27	0.16	0.15		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+45

Nomenclature: See separate page